Lesson 12: Capstone: Algorithm App

Module 4 | 90 minutes | Capstone project | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. pseudocode ___ A. Turning an algorithm into working code.
b. efficiency ___ B. How the work grows with the input.
c. implementation ___ C. An algorithm written in plain, structured steps.

2. Which algorithm would you use?

Write each item in the right group: Binary search / Bubble sort / Caesar cipher / Recursion.

a. Find a name fast in a sorted contact list → _____
b. Rank players by score → _____
c. Hide a message from classmates → _____
d. Draw a snowflake made of smaller snowflakes → _____
e. Put words in alphabetical order → _____

3. My plan

Plan your project: Build an app around a classic algorithm (search, sort, cipher or recursion): pseudocode first, the algorithm in a procedure, input and validation, clear output, step counts for several inputs, and a Python version of the algorithm. Sketch it and list what each part will do.

4. Checklist

Tick each one when your project does it.

a. ☐ Pseudocode
b. ☐ Procedure
c. ☐ Nested loops or recursion
d. ☐ Input and validation
e. ☐ Clear output
f. ☐ Step counts
g. ☐ Python version

5. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ My app is built around an algorithm in a procedure.

☐ I measured its efficiency.

☐ I wrote the Python version and presented it.